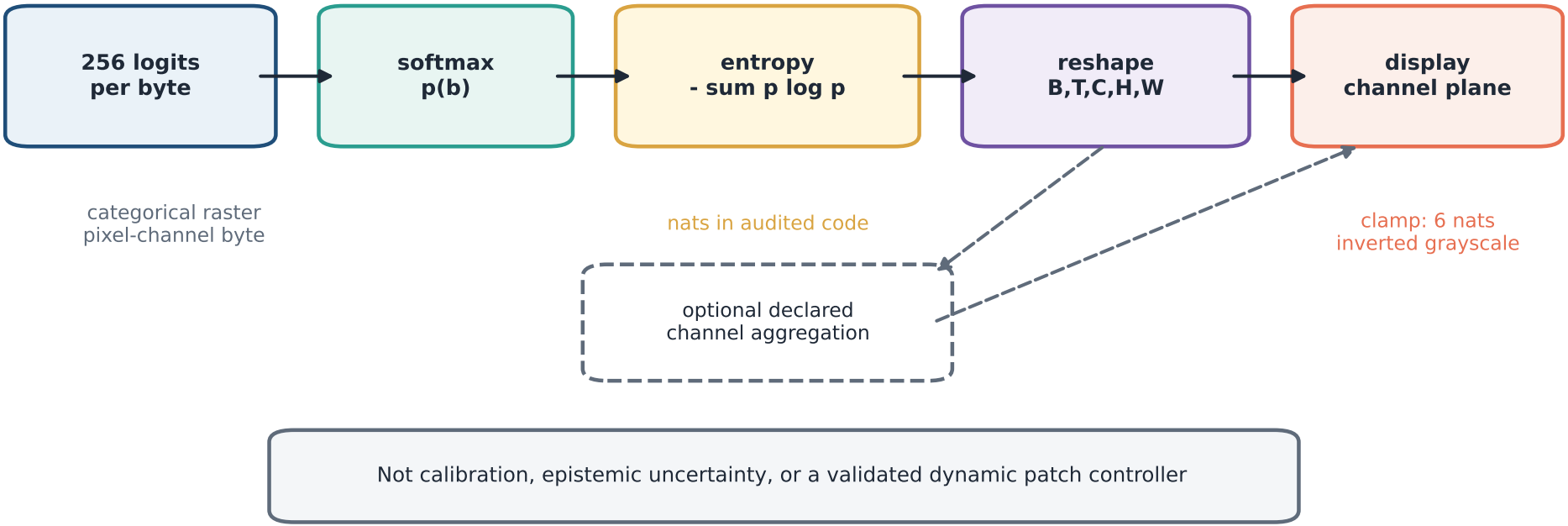Categorical byte predictions expose a spatial dispersion diagnostic